OS – Jackfruit Problem

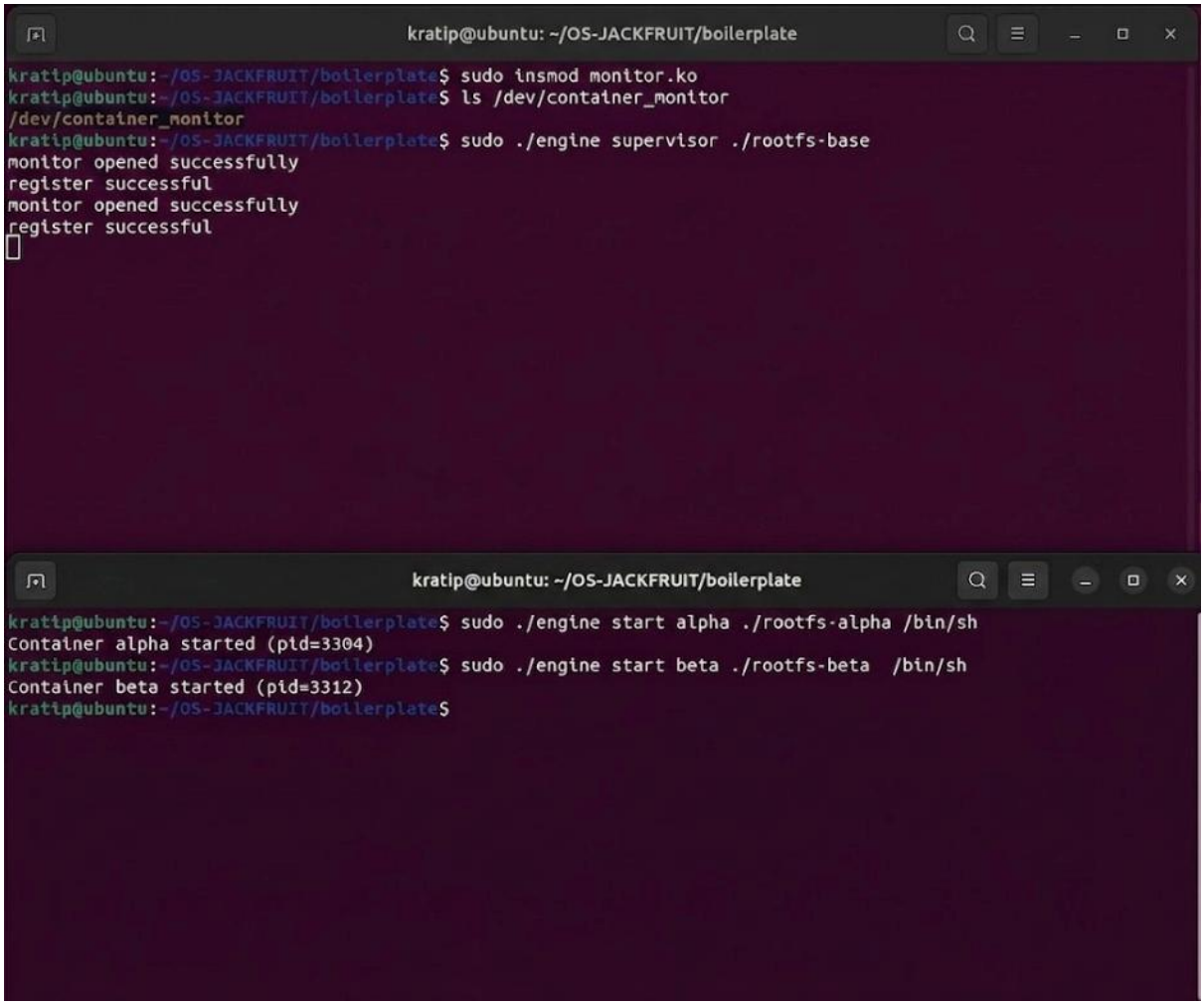
Jashruth K A – PES2UG24AM069

Krati Patel – PES2UG 24AM077

Submission:

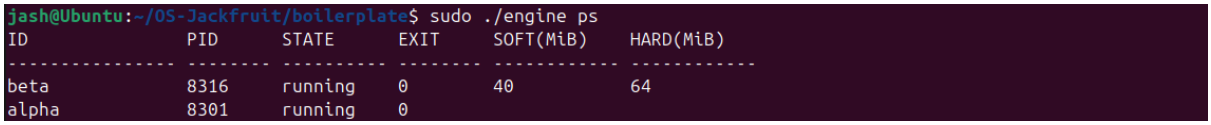
Screenshot 1:

Multi-container supervision - Two containers (alpha and beta) running under one supervisor process



Screenshot 2:

Metadata Tracking - Output of `engine ps` command showing container IDs, PIDs, STATE, EXIT code, and memory limits.



```
jash@Ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
ID               PID      STATE      EXIT     SOFT(MiB)    HARD(MiB)
---------------- -------- ---------- -------- ------------ ------------
beta             8316     running    0        40           64
alpha            8301     running    0
```

Screenshot 3:

Bounded-Buffer Logging - Log file contents captured through logging pipeline (producer-consumer buffer)

```
jash@Ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine logs log1
Log for 'log1':
Message
Message
Message
Message
Message
First log message
First log message
First log message
```

Screenshot 4:

CLI and IPC - CLI command `engine start test123` and supervisor responding with "IPC Working"

```
jash@Ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine start test123 ./rootfs-base "echo IPC Working"
Started container 'test123' pid=6832
jash@Ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
ID               PID      STATE      EXIT     SOFT(MiB)    HARD(MiB)
---------------- -------- ---------- -------- ------------ ------------
test123          6832     exited     0        40           64
log2             6643     exited     0
jash@Ubuntu:~/OS-Jackfruit/boilerplate$ sudo ./engine logs test123
Log for 'test123':
IPC Working
```

Screenshot 5:

Soft-limit warning - `dmesg` output showing soft-limit warning when container exceeds 20 MiB

```
kratip@ubuntu:~/OS-JACKFRUIT/boilerplate$ sudo ./engine start mem ./rootfs-alpha ./memory_hog --
soft-mib 20 - -hard-mib 40
Container mem started (pid=3679)
kratip@ubuntu:~/OS-JACKFRUIT/boilerplate$ sudo dmesg | grep -E "SOFT"
[  3185.497437] [container_monitor] SOFT LIMIT container=mem pid=3647 rss=25821184 limit=20971520
[  3271.191950] [container_monitor] SOFT LIMIT container=mem pid=3665 rss=25821184 limit=20971520
[  3333.848624] [container_monitor] SOFT LIMIT container=mem pid=3679 rss=25821184 limit=20971520
kratip@ubuntu:~/OS-JACKFRUIT/boilerplate$ sudo dmesg | grep -E "HARD"
```

Screenshot 6:

Hard-limit Enforcement - `dmesg` output showing container killed after exceeding 40 MiB hard limit

```
kratip@ubuntu:~/OS-JACKFRUIT/boilerplate$ sudo ./engine start mem ./rootfs-alpha ./memory_hog --
soft-mib 20 --hard-mib 40
Container mem started (pid=3679)
kratip@ubuntu:~/OS-JACKFRUIT/boilerplate$ sudo dmesg | grep -E "SOFT"
[ 3185.497437] [container_monitor] SOFT LIMIT container=mem pid=     rss=25821184 limit=20971520
[ 3271.191950] [container_monitor] SOFT LIMIT container=mem pid5     rss=25821184 limit=20971520
[ 3333.848624] [container_monitor] SOFT LIMIT container=mem pid5     rss=25821184 limit=20971520
kratip@ubuntu:~/OS-JACKFRUIT/boilerplate$ sudo dmesg | grep -E "HARD"
[ 3187.573532] [container_monitor] HARD LIMIT container=mem pid=3647 rss=42598400 limit=41943040
[ 3335.976365] [container_monitor] HARD LIMIT container=mem pid=3679 rss=42598400 limit=41943040
kratip@ubuntu:~/OS-JACKFRUIT/boilerplate$
```

Screenshot 7:

Scheduling experiment - `ps` output with `R<+` (high priority) and `RN+` (low priority) CPU usage difference

```
root         7093 83.9  0.0   1004   584 pts/3    RN+  16:10  27:11 /cpu_hog
root         7106 98.1  0.0   1004   584 pts/3    R<+  16:11  30:46 /cpu_hog
root         7579  6.5  0.0   1004   584 pts/4    RN+  16:38   0:17 /cpu_hog
root         7588 90.9  0.0   1004   584 pts/4    R<+  16:38   3:57 /cpu_hog
vboxuse+     7846  0.0  0.0   9144  2248 pts/2    S+   16:43   0:00 grep --color=auto cpu_hog
```

Screenshot 8:

Clean teardown - `ps aux | grep defunct` showing no zombie processes after shutdown

```
kratip@ubuntu:~/OS-JACKFRUIT/boilerplate$ sudo ./engine stop alpha
Container alpha stopped
kratip@ubuntu:~/OS-JACKFRUIT/boilerplate$ ps aux | grep defunct
kratip        4136  0.0  0.0  9080  2636 pts/1   S+   00:26   0:00 grep --color=auto defunct
kratip@ubuntu:~/OS-JACKFRUIT/boilerplate$ sudo ./engine stop beta
kratip@ubuntu:~/OS-JACKFRUIT/boilerplate$ ps aux | grep defunct
kratip       4141  0.0  0.0  9080 2632 pts/1  S+   00:26   0:00 grep --color=auto defunct
kratip@ubuntu:~/OS-JACKFRUIT/boilerplate$
```